

STE-ANNE-DE-BELLEVUE, QC, Canada

WMO: 713770

Lat: 45.4272N

Lon: 73.9292W

Elev: 128

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-5.5	-19.9	1.9	-9.0	-15.0	2.4	-4.7	23.5	29.3	21.2	23.8	7.5	250	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	86.1	71.4	83.4	69.8	80.8	68.4	74.3	82.3	72.5	79.8	70.9	77.6	9.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.7	117.8	78.7	70.1	111.2	76.6	68.4	105.0	75.1	37.9	82.3	36.2	79.8	34.8	77.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	17.3	15.4	-16.6	90.5	5.2	2.3	-20.3	92.2	-23.3	93.6	-26.2	94.9	-29.9	96.5		
			-17.2	77.6	4.9	2.1	-20.7	79.1	-23.5	80.3	-26.3	81.5	-29.8	83.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	15.7	18.3	28.4	43.0	56.6	65.6	70.1	68.1	60.7	48.5	35.9	23.2	
	DBStd	21.02	12.67	11.00	10.47	8.29	7.55	6.56	4.97	5.32	7.24	7.69	8.84	11.03	
	HDD50	4270	1063	888	673	239	22	0	0	0	4	123	428	830	
	HDD65	7871	1528	1308	1136	661	277	71	11	29	168	513	874	1295	
	CDD50	2317	0	0	2	29	227	467	623	562	326	77	4	0	
	CDD65	443	0	0	0	1	18	88	169	126	40	1	0	0	
	CDH74	3164	0	0	1	12	188	682	1171	817	284	9	0	0	
	CDH80	722	0	0	0	1	40	170	278	172	61	0	0	0	
Wind	WSAvg	7.3	8.2	8.1	8.2	8.3	7.6	6.8	6.4	5.9	6.1	7.1	7.7	7.8	
Precipitation	PrecAvg	39.4	2.9	2.4	2.6	3.4	3.4	3.8	3.8	3.5	3.5	3.8	3.1	3.5	
	PrecMax	51.8	5.0	4.8	4.6	5.9	6.7	6.3	5.9	7.6	6.8	8.0	5.5	6.3	
	PrecMin	32.3	1.2	1.0	0.8	0.6	1.4	2.2	1.9	0.9	1.1	1.2	0.8	0.9	
	PrecStd	4.6	1.0	0.8	0.9	1.5	1.3	1.3	1.1	1.4	1.3	1.8	1.0	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	45.2	60.0	75.2	84.6	88.8	89.8	88.4	86.5	75.1	61.9	50.9	
		MCWB	45.1	40.9	50.5	60.2	66.9	72.8	74.3	72.9	71.6	64.2	55.6	48.5	
	2%	DB	41.1	40.4	50.9	67.2	79.2	84.8	85.8	84.2	80.8	69.4	56.7	44.8	
		MCWB	38.6	37.1	43.8	53.2	63.4	69.7	71.9	70.6	69.3	60.9	52.6	42.6	
	5%	DB	37.3	37.2	45.5	61.3	74.6	81.2	83.0	81.4	77.0	64.4	52.5	40.2	
		MCWB	35.4	34.8	39.7	49.8	59.9	67.6	70.6	69.1	66.9	57.5	48.3	38.2	
	10%	DB	33.6	34.5	41.7	56.3	70.3	77.8	80.3	78.6	73.1	60.2	48.4	36.8	
		MCWB	32.1	32.5	37.0	47.0	58.7	65.7	68.7	68.1	65.1	54.7	44.5	34.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.4	42.2	53.7	61.9	70.0	75.6	77.6	75.7	74.0	66.5	57.3	49.0
			MCDB	46.2	44.1	57.7	72.4	80.2	84.8	87.0	84.3	82.2	72.2	59.7	51.0
2%		WB	39.4	37.7	44.9	55.9	66.5	72.7	74.8	73.2	71.0	61.9	52.9	42.7	
		MCDB	40.6	39.7	49.9	64.1	74.5	80.9	82.1	81.2	77.5	67.4	56.3	44.6	
5%		WB	35.5	35.0	40.4	51.5	63.5	70.4	72.8	71.4	68.7	58.5	48.9	38.4	
		MCDB	37.3	36.9	44.5	59.0	70.9	77.6	79.7	77.8	75.1	63.5	52.1	40.0	
10%		WB	32.3	32.7	37.6	48.1	60.4	68.2	71.0	69.8	66.3	55.3	44.8	35.0	
		MCDB	33.7	34.5	40.9	55.1	67.3	75.2	77.8	75.8	72.0	59.7	47.8	36.5	
Mean Daily Temperature Range		5% DB	MDBR	14.6	15.5	15.0	16.7	18.5	17.6	17.0	17.5	18.1	14.9	12.4	12.1
			MCDBR	17.1	15.6	19.0	26.3	25.5	22.2	19.9	20.4	22.0	21.4	19.0	14.0
	MCWBR		16.4	14.2	13.7	15.6	12.8	10.9	9.9	9.7	11.8	13.5	15.5	13.1	
	5% WB	MCDBR	16.3	15.0	17.7	22.8	20.7	18.4	17.2	17.3	19.6	19.3	18.3	13.7	
		MCWBR	15.8	14.2	14.0	15.3	12.7	10.8	10.1	9.7	11.9	13.5	15.8	13.1	
Clear-Sky Solar Irradiance	taub	0.261	0.289	0.329	0.362	0.387	0.443	0.430	0.414	0.374	0.338	0.307	0.289		
	taud	2.351	2.267	2.306	2.364	2.351	2.198	2.245	2.284	2.383	2.490	2.489	2.325		
	Ebn at Noon	272	283	285	285	279	263	265	265	267	264	253	244		
	Edh at Noon	23	31	35	36	38	45	42	39	33	25	21	22		
All-Sky Solar Radiation	RadAvg	452	756	1155	1397	1630	1783	1803	1595	1258	732	446	320		
	RadStd	52	75	96	155	158	137	140	75	103	65	55	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page